

Oline A. Ranum

Cognitive Vision Lab, Centre for Vision, Speech and Signal Processing (CVSSP) · University of Surrey
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- Researcher targeting **an inclusive future for AI technology**.
- **PhD researcher** in **sign language modelling & AI** at the **University of Surrey** (CVSSP).
- **Master's degree** in **Artificial Intelligence** with ELLIS Honours & Cum Laude.
- **FNWI Master's Thesis Prize 2025** (University of Amsterdam) & **Amsterdam AI Thesis Award 2024**.
- Founder of **DeepSign AS** & **BliFlink** NGO.

Education

Doctor of Philosophy (PhD), University of Surrey (CVSSP), UK Jul. 2025 – present

- PhD in Sign Language Modeling and AI with the Cognitive Vision Lab.
- Sign language recognition and translation, computer vision, multimodal learning.

Master of Science (MSc), Artificial Intelligence, University of Amsterdam 2022 – 2024

- **ELLIS Honours & Cum Laude** | 120 EC.
- **Thesis: *3D Awareness, Geometry, and Linguistics in Neural Sign Language Representations***.
- Recipient of the FNWI Master's Thesis Prize and the Amsterdam AI Thesis Award.
- Geometric deep learning, computer vision, natural language processing.

Master of Science (MSc), Astroparticle and Gravitational Physics, University of Amsterdam 2020 – 2021

- Partial completion | 54 EC – entrepreneurship, detector engineering, mathematics for physics.

Bachelor of Science (BSc), Physics and Astronomy, University of Oslo 2016 – 2020

- 210 EC – computational physics, mathematics, programming.

Bachelor of Science (BSc), International Relations, University of Oslo 2015 – 2016

- Partial completion | 60 EC – international law, human rights, political theory, philosophy.

Study abroad, University of California, Berkeley (UCB), USA | 30 EC – data-driven astronomy, nuclear engineering 2019

Summer school, Stellenbosch University, South Africa | 5 EC – *Advanced Nuclear Reactions and Applications in Astrophysics* (INTPART) 2019

Publications & Preprints

- [1] Ranum, O., Hadfield, S., Bowden, R. (2026). *What's the Point? Spatial Grammar & Index Resolution for Sign Language Recognition*. arXiv preprint. arxiv.org/abs/2606.08056
- [2] Cory, O., Ivashechkin, M., Sahin, K., Ranum, O., Low, J., Fish, E., Pelykh, A., Mercanoglu Sincan, O., Bowden, R. (2026). *BackTranslation2.0 – A Linguistically Motivated Metric to Assess Sign Language Production*. European Conference on Computer Vision (ECCV) 2026 (*accepted*). arxiv.org/abs/2606.28673
- [3] Brown, M., Ranum, O., Fish, E., Proctor, H., Woll, B., Bowden, R., Cormier, K. (2026). *SignGPT and the Visual Language Toolkit*. Proceedings of the LREC 2026 12th Workshop on the Representation and Processing of Sign Languages: Language in Motion. sign-lang.uni-hamburg.de/lrec/pub/26026.pdf
- [4] Mercanoglu Sincan, O., Pelykh, A.^{*}, Fish, E.^{*}, Walsh, H.^{*}, Low, J.^{*}, Sahin, K.^{*}, Ranum, O.^{*}, Asasi, S.^{*}, Emery, S.^{*}, Bowden, R. (2027). *Machine Translation for Sign Languages*. Annual Review of Linguistics, 2027 (*accepted*).
^{*}Equal contribution; preprint not yet available.

- [5] Ranum, O., Wessels, D., Otterspeer, G., Bekkers, E., Roelofsen, F., Andersen, J. (2024). *The NGT200 dataset — geometric multi-view isolated sign recognition*. ICML 2024 Workshop on Geometry-grounded Representation Learning and Generative Modeling. openreview.net/forum?id=idkNzTC67X
- [6] Ranum, O., Otterspeer, G., Andersen, J., Roelofsen, F. (2024). *3D-LEX v1.0: 3D Lexicons for American Sign Language and Sign Language of the Netherlands*. Proceedings of the LREC-COLING 2024 11th Workshop on the Representation and Processing of Sign Languages: Evaluation of Sign Language Resources, pages 252–263, Torino, Italy. ELRA and ICCL. aclanthology.org/2024.signlang-1.33/
- [7] Hu, A., Ranum, O., Pozrikidou, C., Zhou, M. (2023). *Reproducibility study of “Joint Multisided Exposure Fairness for Recommendation”*. ML Reproducibility Challenge 2022. openreview.net/forum?id=A0Sjs3IJWb-
- [8] Rienäcker, B., Brusa, R. S., Caravita, R., Mariazzi, S., Penasa, L., Pino, F., Ranum, O., Nebbia, G. (2022). *A fiber detector to monitor ortho-Ps formation and decay*. Nuclear Instruments and Methods in Physics Research Section A, 1027, 166275. sciencedirect.com/science/article/pii/S0168900221011062

Research Experience

Research Assistant, Cognitive Vision Lab, University of Surrey, Guildford, UK Jul. 2025 – present

- Data capture and research software management.

Researcher, SignLab Amsterdam, Netherlands 2024

- Automatic annotation of visual phonetics; interface development for linguistic research.

Research Projects, University of Amsterdam, Netherlands 2023 – 2024

- *Sign Language Processing*: [5]; [6]; Hand shape classification.
- *Deep Learning*: SE(3)-equivariant neural rendering.
- *Computer Vision*: steerable diffusion for scene generation.
- *Engineering*: assistive speaker watch for individuals with communicative challenges.

Research Project, DeepSign AS, Oslo, Norway 2022

- Machine translation methods for Norwegian and Norwegian Sign Language.

Technical Student & Research Intern, CERN, AEgIS Group, Geneva, Switzerland 2019 – 2020

- Software for analyzing positronium production; detector calibration [8].

Research Intern, Lawrence Berkeley National Lab (LBNL) & Berkeley Astrophysical Institute, USA 2019

- *Nuclear Physics* (Bernstein Group): detector calibration; analysis of Yt86 production.
- *Astrophysics*: classification of white dwarf spectra using the Cannon model.

Research Assistant & Lab Instructor, University of Oslo, Norway 2018

- Laboratory assignments and teaching for undergraduate physics courses; RGB-based motion capture system for chaotic motion detection.

Awards, Grants & Honours

- **Master’s Thesis Prize**, Faculty of Science (FNWI), University of Amsterdam 2025
- **Amsterdam AI Thesis Award**, best AI thesis in Amsterdam 2024
- **ELLIS Honours**, European Laboratory for Learning and Intelligent Systems | EUR 4,000 2023
- **STUD-ENT Grant**, Norwegian Research Council, DeepSign project | EUR 100,000 2021

Entrepreneurship & Industry

Data Annotator and Quality Controller, Accenture, Amsterdam, Netherlands 2025

- Quality control of data for training LLMs at Apple Inc.

- CEO & Founder**, DeepSign AS, Oslo, Norway 2021
- Company management, strategy, finances, sales, team development, and product development.
 - Accelerators: [ACE Cohort 2022](#), [Demonstrator Lab](#); media coverage: [Titan.uio.no](#), [Argument.no](#).
- CTO & Founder**, BliFlink, Oslo, Norway 2020
- Online tutoring platform offering free educational resources during COVID-19.
 - Management, implementation, and product development.

Talks & Academic Service

Invited Talks

- “Student entrepreneurship and innovation for social good”
- [University of Oslo](#), “[A New Beginning](#)”, UiO, Norway Aug. 2021
- “Diagnosing positronium using the fast annihilation cryogenic tracker of AEGIS at CERN”
- AEGIS Collaboration Meeting, CERN, Switzerland Oct. 2019
 - INTPART, Witwatersrand University, South Africa Nov. 2019

Supervision, Representation & Outreach

- Supervisor, Bachelor’s thesis in Business Development and Innovation, VU Amsterdam 2020
- Student representative, Physics Student Council, UiO: Board of Education; Institute Board 2017 – 2018
- Founder and organizer, physics discussion groups, UiO, Norway 2018
- Event manager and organizer, IAESTE Career Fair, Norway 2017 – 2018
- Representative, SAIH – Students and Academics International Help Foundation, UiO 2015

Skills & Languages

Programming

Python (10+ years) · C++ (2 years) · Matlab (4 years) · ADQL/SQL (1 year)

Machine Learning & Vision

Domains: sign language recognition, translation, and linguistics; computer vision; deep learning

Frameworks: PyTorch · TensorFlow · OpenCV · Geometric Deep Learning; Operating systems: Linux · Windows

Research & Professional

Research: scientific writing, report writing, background research, research planning, SL data collection

Soft skills: excellent communication, organizational, and problem-solving skills; fast learner

Project management: financial planning, coordination of research activities

Languages

Norwegian (native) · English (fluent) · Dutch (beginner)

Norwegian Sign Language (Basic working knowledge) · British Sign Language (Level 1) · Dutch Sign Language (beginner)